1. Cover Page

AIVI Career Intelligence OS

Complete Professional Project Documentation

AI-powered career intelligence, resume optimization, job-fit analysis, and professional growth tooling.

Field	Value
Project name	AIVI Career Intelligence OS
Version	0.0.0
Date	31 May 2026
Author/team	AIVI Labs / Codex Documentation Pass
Repository	C:\Users\abhis\OneDrive\Documents\GitHub\aivi-career
Primary runtime	React/Vite frontend plus Firebase Auth, Firestore, and Cloud Functions

Audience: developers, investors, clients, operators, and new team members who need a complete understanding of the product, codebase, runtime architecture, and current gaps.

2. Table of Contents

1. Cover Page

2. Table of Contents

3. Project Overview

4. Product Requirements

4A. Pricing & Plans

5. User Flows & Navigation

6. System Architecture

7. Tech Stack

8. All External APIs & Services Used

9. Internal API Reference

10. Database Schema

11. Authentication & Authorisation

12. Rate Limiting & Security

13. Background Jobs & Workers

14. Environment Variables

15. Local Setup Guide

16. Testing

17. Deployment Guide

18. Known Issues & TODOs

19. Glossary

3. Project Overview

AIVI Career Intelligence OS is a client-side React application backed by Firebase services that helps users understand, improve, and reposition their professional profile. It combines resume scoring, job-fit analysis, LinkedIn analysis, resume generation, skill-roadmap planning, interview preparation, promotion readiness, portfolio evaluation, and admin analytics into one guided dashboard.

The product solves a common career problem: candidates often have scattered documents, unclear positioning, weak ATS alignment, and no structured way to decide what to improve next. AIVI turns raw resume/profile/project inputs into structured intelligence, scores, action plans, and generated career assets.

It is built for job seekers, freshers, mid-career professionals, promotion candidates, and operators who need a scalable career-intelligence workflow. Its main difference from generic AI tools is that prompts, scoring, quota control, output schemas, and user journeys are productized rather than left to open-ended prompting.

Core Value Proposition	Explanation
Career visibility	Improve how a candidate appears to ATS systems, recruiters, hiring managers, and internal promotion stakeholders.
Structured intelligence	Return schema-driven analyses, scores, and action plans instead of generic text.
Workflow continuity	Connect resume, job fit, skill gaps, interview prep, LinkedIn, portfolio, and promotion planning.
Cost-controlled AI	Plan gating, lifetime counters, monthly counters, caching, token logging, and trial limits protect AI spend.

4. Product Requirements

The current product includes public marketing pages, authentication, a protected dashboard, AI analysis modules, trial/plan gating, admin operations, contact capture, and document/PDF utilities.

Feature / Module	Entry Point	What It Does	Why It Exists
Landing and pricing	/	Explains AIVI, product modules, pricing anchor, and CTAs.	Converts visitors into sign-ups and explains the product category.
Authentication	/auth, /login	Email/password sign-in, sign-up, password reset, Google popup sign-in, email verification send.	Creates persistent Firebase-authenticated user accounts.
Dashboard home	/dashboard	Workspace overview, plan usage, recent results, trial activation prompt.	Central command center for returning users.
Resume Intelligence	/dashboard/resume	Accepts pasted resume or PDF, runs resume analysis, shows scores and guidance.	Finds ATS, keyword, clarity, formatting, leadership, and quantification gaps.
Resume Studio	/dashboard/builder	Builds or tailors resume drafts from existing resume, target role, JD, and fresher profile inputs.	Turns analysis into a usable generated resume asset.
Job Fit Analysis	/dashboard/job-fit	Compares resume against a JD; computes deterministic match scores.	Guides whether to apply and what to change before applying.
LinkedIn Analyzer	/dashboard/linkedin	Analyzes LinkedIn profile text and optionally generates optimized copy.	Improves recruiter search visibility and credibility.
Skill Roadmap	/dashboard/roadmap	Builds 30/60/90-day skill plan from missing skills and optional resume context.	Turns gaps into learning actions.
Interview Prep	/dashboard/interview	Generates categorized interview questions, key points, mistakes, and sample answers.	Prepares users for role-specific interview loops.
Promotion Readiness	/dashboard/promotion	Compares current role to target role and returns readiness score, gaps, plan, and timeline.	Supports internal advancement and negotiation.
Portfolio Analyzer	/dashboard/portfolio	Reviews project links and project descriptions for depth and attractiveness.	Aligns public project proof with target roles.
Settings	/dashboard/settings	Profile display name update, password update, subscription view, FIRST1000 coupon, notification toggles.	Lets users manage identity and access state.

Admin Dashboard	/dashboard/admin	User list, plan/role edits, quota reset, usage/cost analytics, contact messages.	Provides operational control and business visibility.
Contact	/contact	Collects name, email, subject, message and saves/sends notification.	Inbound support and sales lead capture.

4.1 End-to-End User-Facing Flows

1. Visitor lands on /, opens Product or AI Tools navigation, reviews pricing, and chooses Sign In or Enter AIVI.
2. New user creates an account with email/password or Google. Email/password sign-up optionally stores displayName and sends an email verification.
3. ProtectedRoute allows dashboard only when Firebase Auth has a user; unauthenticated users are redirected to /auth.
4. Dashboard reads user plan and recent analysis history from Firestore, then presents module cards and plan/trial prompts.
5. A user enters or uploads content. PDF upload calls extractPdf; pasted text goes directly to the module flow.
6. The UI calls checkUsage for a fast preflight. The AI callable performs the authoritative quota increment and rejects if needed.
7. The Firebase Function builds a Gemini prompt, calls Gemini, stores result and usage logs, and returns structured JSON.
8. The page renders scores, guidance, charts, exports, history, and next actions. If blocked, UpgradeModal points to pricing; FIRST1000 can activate a trial.

4A. Pricing & Plans

This section was added after reviewing the active landing page components, frontend copy, Firebase/Firestore plan logic, config files, payment-related routes, package dependencies, and hardcoded pricing values. The currently working commercial model is freemium plus a beta coupon/manual admin plan assignment; Stripe and Razorpay are not live integrations in the codebase.

4A.1 Pricing Tiers

Plan	Price	Included	Limits	Designed For
Free / Discovery	₹0 forever. No annual price configured.	Landing copy: try all 6 modules, limited lifetime access, Resume and Job Match preview, basic score breakdown, and ability to see the core AIVI workflow before upgrading.	Backend limits: resume 2 lifetime, job_fit 2 lifetime, roadmap 1, interview 1, promotion 1, portfolio 1, linkedin_analyze 2, resume_rebuild 0, linkedin_optimize 0. No paid API-call allowance exists.	New users validating the product before paying.
Starter	₹399/month. No annual price configured.	Landing copy: 30 analyses per month, full Skill Roadmap, Interview Intelligence, Promotion Readiness, Portfolio Analyzer, Resume Rewrite, Version tracking, and all modules. Beta badge: 30-Day Free Trial for First 1000 Users.	30 analyses per rolling 30-day reset window. Trial users get Starter access for 30 days through FIRST1000, capped at first 1000 redemptions.	Active job seekers who need repeated tactical optimization.
Pro	₹599/month. No annual price configured.	Landing copy: unlimited analyses, priority AI processing, advanced career insights, export reports (PDF), Mock Interview future-ready, premium support, and everything in Starter with deeper optimization.	Server treats Pro as unlimited for gating and increments monthly usage only for analytics. Some advertised items, such as priority processing and mock-interview future-ready support, are copy/product promises rather than separate payment-provider logic.	Power users, heavy job seekers, and mid-senior professionals who need repeated analysis.
Enterprise	Not applicable.	No Enterprise tier, pricing card, route, plan enum, or backend gating branch was found.	Not applicable.	Not currently offered by this codebase.
Admin	Internal only; not sold publicly.	Admin role bypasses normal usage checks and can manage user roles/plans from the admin dashboard.	Unlimited by role; not a customer subscription tier.	Operators and platform administrators.

4A.2 Feature Comparison Table

Feature / Limit	Free	Starter	Pro	Admin
Resume analysis	Limited: 2 lifetime	Included: counts toward 30/month	Included: unlimited	Included: unlimited
Job Fit analysis	Limited: 2 lifetime	Included: counts toward 30/month	Included: unlimited	Included: unlimited
Skill Roadmap	Limited: 1 lifetime	Included: counts toward 30/month	Included: unlimited	Included: unlimited
Interview Intelligence	Limited: 1 lifetime	Included: counts toward 30/month	Included: unlimited	Included: unlimited
Promotion Readiness	Limited: 1 lifetime	Included: counts toward 30/month	Included: unlimited	Included: unlimited
Portfolio Analyzer	Limited: 1 lifetime	Included: counts toward 30/month	Included: unlimited	Included: unlimited
LinkedIn analysis	Limited: 2 lifetime	Included: counts toward 30/month	Included: unlimited	Included: unlimited
Resume Rewrite / Rebuild	Excluded	Included	Included	Included
LinkedIn optimization	Excluded	Included	Included	Included
Version tracking	Limited by feature access	Included per landing copy	Included	Included
PDF report export	Not advertised	Not advertised	Included per landing copy	Operational/admin access depends on feature implementation
Priority AI processing	Excluded	Excluded	Advertised, but no separate provider/billing implementation found	Not applicable
Premium support	Excluded	Excluded	Included per landing copy	Not applicable
Payment self-service	Excluded	Not implemented	Not implemented	Manual plan changes available

4A.3 Payment & Billing Flow

Upgrade CTAs currently route users to the landing page pricing anchor or dashboard rather than to a provider checkout. The active sources are the landing pricing cards, UpgradeModal, PlanUsageBanner, Settings subscription panel, Terms page, and Admin Dashboard plan controls.

Area	Current Working Logic
Provider	No live Stripe or Razorpay integration was found. package.json and functions/package.json do not include Stripe/Razorpay SDKs, and no checkout, order creation, payment verification, billing portal, or subscription-management endpoints exist.
Upgrade path	Pricing cards link to /dashboard. UpgradeModal links to /#pricing and emits a gtag upgrade_clicked event when available. Settings links users back to /#pricing for Starter or Pro.
Plan activation	Automatic activation exists only through applyCoupon(FIRST1000), which sets plan_type to starter and trial=true. Admins can manually set planType/plan_type through the admin-only updateUserRole callable.
Billing cycle	Terms copy says paid plans are monthly and may be cancelled anytime with access until the end of the billing period, but there is no provider-backed subscription record or renewal enforcement in the code.
Payment failure	Not applicable today: no payment is attempted, so there is no failed-payment retry, dunning, grace period, invoice_failed handler, or automatic downgrade tied to billing failure.

4A.4 Free Trial or Freemium Logic

All new Firebase Auth users receive a Free / Discovery plan document at signup. The beta trial is activated only by coupon code FIRST1000, limited to the first 1000 users, and grants Starter access for 30 days without collecting payment information.

```
// Working limits from planService.ts and functions/src/index.ts
Free: resume=2, job_fit=2, roadmap=1, interview=1, promotion=1, portfolio=1,
      linkedin_analyze=2, resume_rebuild=0, linkedin_optimize=0
Starter trial: FIRST1000 -> plan_type='starter', trial=true, trialExpiresAt=now+30 days
Starter paid/manual: 30 analyses per month
Pro/manual: unlimited gating; monthly usage still increments for analytics
```

4A.5 Upgrade & Downgrade Behaviour

Event	Behaviour
Free to Starter trial	applyCoupon validates auth, code FIRST1000, trialUsed=false, and system_stats/beta.first1000Used < 1000 in a Firestore transaction. It sets plan_type/planType to starter, trial=true, trialExpiresAt, trialUsed=true, and couponCode.
Free/Starter to Pro	No self-serve payment upgrade exists. An admin can manually set planType/plan_type to pro through updateUserRole.
Starter monthly reset	checkUsageInline resets monthly_analysis_used to 0 when monthly_reset_date is older than 30 days, then increments usage.
Trial expiry / downgrade	During server-side usage, expired trials are changed back to free and trial fields are cleared. Saved user data and prior analysis records remain; future premium/gated actions are denied until plan access is restored.
Manual downgrade	Admins can set planType/plan_type to free or starter. Existing analysis data is not deleted by this function; access is controlled by future usage checks.
Quota reset	Admin resetQuota clears monthly and several lifetime counters. The reset logic currently resets resume, jobfit, roadmap, interview, promotion, and portfolio counters.

4A.6 Webhooks & Subscription Events

Not applicable in the current implementation. The app does not listen to Stripe or Razorpay webhooks and has no routes/functions for payment_succeeded, payment_failed, invoice_failed, subscription_created, subscription_updated, subscription_cancelled, payment.captured, refund, or chargeback events.

Event	Implemented?	Current App Response
Stripe checkout.session.completed	No	No handler exists.
Stripe invoice.payment_succeeded	No	No handler exists.
Stripe invoice.payment_failed	No	No handler exists.
Stripe customer.subscription.deleted/cancelled	No	No handler exists.
Razorpay payment.captured	No	No handler exists.
Razorpay payment.failed	No	No handler exists.
Razorpay subscription events	No	No handler exists.

4A.7 Pricing Page Copy

Exact active landing-page copy from src/pages/LandingConceptB.tsx:

```
Pricing section eyebrow: Fair pricing. Big impact.
Pricing section headline: Plans for every ambition

Free
Price: ₹0 forever
Detail: Discovery
Features:
- Try all 6 modules
- Limited lifetime access
- Resume and Job Match preview
- Basic score breakdown
- See core AIVI workflow before upgrading
CTA: Start Free

Starter
Price: ₹399 /month
Detail: 30-Day Free Trial for First 1000 Users
Badge: Beta Trial
Features:
- 30 analyses per month
- Full Skill Roadmap
- Interview Intelligence
- Promotion Readiness
- Portfolio Analyzer
- Resume Rewrite
- Version tracking
- Built for active job seekers
CTA: Start Starter Plan

Pro
Price: ₹599 /month
Detail: Advanced
Features:
- Unlimited analyses
- Priority AI processing
- Advanced career insights
- Export reports (PDF)
- Mock Interview (future-ready)
- Premium support
- Everything in Starter, with deeper optimization
CTA: Go Pro
```

Related frontend copy: PromoBanner says, 'Beta access' and 'First 1000 users receive full-platform early access with code FIRST1000'. UpgradeModal says, 'You've used your free career insights', 'Upgrade to Starter for unlimited career growth tools and full AI analysis

across all modules.', 'Upgrade to Starter — ₹399/mo', 'Compare Plans', and 'Limited Beta Trial. No payment info required.' Settings says, 'Internal trial system. No payment information required.'

Guarantee / cancellation copy: the pricing cards do not show a money-back guarantee or cancel-anytime footnote. The Terms page states subscription fees are billed monthly, non-refundable except where required by law, pricing may change with 30 days' notice, and users may cancel at any time with access continuing until the end of the current billing period.

5. User Flows & Navigation

Routing is declared in src/App.tsx using react-router-dom. Public pages are accessible without auth. Dashboard routes are nested inside DashboardLayout and wrapped by ProtectedRoute. The public GlobalHeader has Home, Product, AI Tools, Pricing, About, Contact, Sign In, and Enter AIVI CTAs. The dashboard sidebar exposes module navigation, Settings, Sign Out, and Admin Panel for admins only.

Route	Component	Purpose
/	Landing	Public landing page with value proposition, pricing anchor, CTAs, and marketing sections.
/auth	Auth	Email/password and Google sign-in/sign-up flow.
/login	Auth	Alias for the auth page.
/verify-email	VerifyEmail	Email verification and resend/check controls.
/about	About	Public about page.
/how-it-works	HowItWorks	Public product explanation page.
/what-is-aivi	WhatIsAivi	Public educational page defining AIVI.
/ai-resume-builder	AIResumeBuilderPage	SEO/marketing page for AI resume builder.
/linkedin-optimization	LinkedInOptimizationPage	SEO page for LinkedIn optimization.
/contact	Contact	Contact form backed by Firebase callable submitContactForm.
/privacy	Privacy	Privacy policy.
/terms	Terms	Terms and subscription/billing language.
/ai-resume-builder-for-freshers	AIResumeBuilderFreshers	SEO page for fresher resume builder.
/resume-scoring-software	ResumeScoringSoftware	SEO page for resume scoring software.
/job-fit-analysis-tool	JobFitAnalysisTool	SEO page for job-fit analysis.
/career-gap-analysis-ai	CareerGapAnalysisAI	SEO page for skill gap and career gap analysis.
/ai-vs-chatgpt-vs-gemini	AIVsChatGPTVsGemini	Comparison page positioning AIVI against generic AI tools.
/dashboard	DashboardHome	Protected dashboard overview, plan banner, recent analyses, trial activation.
/dashboard/resume	ResumeIntelligence	Resume analysis with PDF/text input and score guidance.
/dashboard/builder	ResumeBuilder	Resume Studio for rebuild, tailoring, and fresher first-resume flows.
/dashboard/job-fit	JobFitAnalysis	Resume versus job description analysis with deterministic scores.
/dashboard/linkedin	LinkedInAnalyzer	LinkedIn profile analysis and premium optimization.
/dashboard/roadmap	SkillRoadmap	30/60/90-day roadmap from missing skills and resume context.
/dashboard/interview	InterviewPrep	Interview question banks and answer scaffolding.
/dashboard/promotion	PromotionReadiness	Promotion readiness assessment and action plan.
/dashboard/portfolio	PortfolioAnalyzer	Portfolio/project analyzer.
/dashboard/settings	SettingsPage	Profile, password, subscription, coupon, notification preferences.
/dashboard/admin	AdminDashboard	Admin analytics, plan edits, role edits, contact messages.
*	NotFound	Fallback route for unknown paths.

5.1 Complete Primary Journey

9. Awareness: user reaches the landing page or an SEO page such as /job-fit-analysis-tool.
10. Activation: user clicks Enter AIVI or Sign In, then authenticates through Firebase Auth.
11. Workspace entry: user lands on /dashboard and sees plan status, saved intelligence, and module shortcuts.
12. Core action: user selects a module, supplies resume/profile/JD/project data, and starts analysis.
13. Backend execution: callable function verifies auth, checks plan/role, calls Gemini, stores result, logs usage, and returns JSON.

14. Outcome: user reviews scores, reads action plan, exports or copies content, and moves into another connected module.
15. Monetization/trial: when limits are reached, UpgradeModal explains Starter/Pro pricing; FIRST1000 can temporarily unlock Starter.

6. System Architecture

The production architecture is a Firebase-backed SPA. Supabase migrations and Edge Functions are present as legacy migration artifacts, but the active frontend services import Firebase clients and Firebase callable functions.

```
Browser (React/Vite SPA)
  |-- Firebase Auth SDK: email/password, Google popup, persisted local session
  |-- Firestore client SDK: reads own plan/history/profile where rules allow
  |-- Firebase Functions SDK: callable APIs
        |
        v
Firebase Cloud Functions (Node 20, TypeScript)
  |-- request.auth validation from callable context
  |-- Firestore Admin SDK transactions and writes
  |-- Gemini OpenAI-compatible endpoint and @google/generative-ai SDK
  |-- Nodemailer SMTP for contact notifications
        |
        v
Firestore
  |-- profiles, user_plans, user_roles, analysis_results, ai_usage_logs
  |-- system_stats/beta, ai_rate_limits, contact_messages
        |
        v
External services: Gemini API, Firebase Auth, Hostinger SMTP, static hosting/CDN
```

6.1 Data Flow: AI Analysis

16. User submits content from a dashboard module.
17. Frontend calls checkUsage for UI feedback; this is a read-only quota preflight.
18. Frontend calls careerAnalysis, linkedinAnalyze, linkedinOptimize, or extractPdf through Firebase httpsCallable.
19. Cloud Function checks request.auth, validates input, checks role, and uses checkUsageInline for authoritative plan enforcement.
20. For job_fit and LinkedIn analysis, the server computes a SHA-256 content hash and may return a cached result.
21. Server calls Gemini with temperature 0 and schema-heavy prompts, parses JSON, writes analysis_results, writes ai_usage_logs, and returns structured result.
22. Frontend renders cards, charts, tables, exports, and history from the returned JSON and Firestore reads.

6.2 Repository Map

Path	Role

src/App.tsx	Top-level providers and route tree.
src/pages/	Public pages, SEO pages, dashboard modules, admin, auth, legal pages.
src/components/	Reusable UI, headers, sidebars, score visuals, plan banners, modals.
src/components/ui/	shadcn/Radix UI primitives.
src/services/	Firebase callable wrappers and Firestore read helpers.
src/utils/	PDF parser wrapper, deterministic job-fit scoring, resume export utilities.
src/integrations/firebase/client.ts	Firebase client initialization and emulator switching.
src/integrations/supabase/	Generated legacy Supabase client/types, not the active runtime path.
functions/src/index.ts	All current Firebase Cloud Functions and helpers.
supabase/functions/	Legacy Deno Edge Functions using Lovable AI Gateway.
supabase/migrations/	Legacy PostgreSQL schema/RLS/RPC migrations.
firebase.json	Firebase hosting, functions, Firestore, emulator configuration.
firestore.rules	Client database security boundary.
firestore.indexes.json	Composite indexes for analysis and usage queries.
docs/	Existing technical, business, migration, and brand documentation.
scripts/ and functions/*.cjs	Admin/repair/verification scripts for Firebase data.
public/brand	SVG logo and brand assets.

7. Tech Stack

The table lists package-managed technologies and their current declared versions. Many Radix packages are individual primitives used by shadcn/ui components.

Area	Library / Tool	Version	Reason
Frontend dependency	@hookform/resolvers	^3.10.0	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-accordion	^1.2.11	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-alert-dialog	^1.1.14	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-aspect-ratio	^1.1.7	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-avatar	^1.1.10	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-checkbox	^1.3.2	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-collapsible	^1.1.11	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-context-menu	^2.2.15	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-dialog	^1.1.14	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-dropdown-menu	^2.1.15	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-hover-card	^1.1.14	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-label	^2.1.7	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-menubar	^1.1.15	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-navigation-menu	^1.2.13	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-popover	^1.1.14	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-progress	^1.1.7	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-radio-group	^1.3.7	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-scroll-area	^1.2.9	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-select	^2.2.5	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-separator	^1.1.7	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-slider	^1.3.5	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-slot	^1.2.3	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-switch	^1.2.5	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-tabs	^1.1.12	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-toast	^1.2.14	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-toggle	^1.1.9	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-toggle-group	^1.1.10	Runtime, UI, build, test, or server support library.
Frontend dependency	@radix-ui/react-tooltip	^1.2.7	Runtime, UI, build, test, or server support library.
Frontend dependency	@tanstack/react-query	^5.83.0	QueryClient provider and future data caching surface.
Frontend dependency	class-variance-authority	^0.7.1	Runtime, UI, build, test, or server support library.
Frontend dependency	clsx	^2.1.1	Runtime, UI, build, test, or server support library.
Frontend dependency	cmdk	^1.1.1	Runtime, UI, build, test, or server support library.
Frontend dependency	date-fns	^3.6.0	Runtime, UI, build, test, or server support library.
Frontend dependency	docx	^9.6.1	DOCX export support.
Frontend dependency	embla-carousel-react	^8.6.0	Runtime, UI, build, test, or server support library.
Frontend dependency	firebase	^10.12.0	Client SDK for Auth, Firestore, and callable Functions.
Frontend dependency	firebase-admin	^13.6.1	Admin SDK for privileged Cloud Function database access.
Frontend dependency	framer-motion	^12.34.0	Motion and sidebar animation.
Frontend dependency	input-otp	^1.4.2	Runtime, UI, build, test, or server support library.
Frontend dependency	jspdf	^4.2.1	PDF export support.
Frontend dependency	lucide-react	^0.462.0	Icon library.

Frontend dependency	next-themes	^0.3.0	Runtime, UI, build, test, or server support library.
Frontend dependency	react	^18.3.1	Core UI framework for the SPA.
Frontend dependency	react-day-picker	^8.10.1	Runtime, UI, build, test, or server support library.
Frontend dependency	react-dom	^18.3.1	DOM renderer for React.
Frontend dependency	react-helmet-async	^2.0.5	Runtime, UI, build, test, or server support library.
Frontend dependency	react-hook-form	^7.61.1	Runtime, UI, build, test, or server support library.
Frontend dependency	react-resizable-panels	^2.1.9	Runtime, UI, build, test, or server support library.
Frontend dependency	react-router-dom	^6.30.1	SPA route and nested dashboard routing.
Frontend dependency	recharts	^2.15.4	Charts for dashboards and analysis visualizations.
Frontend dependency	sonner	^1.7.4	Toast notifications.
Frontend dependency	tailwind-merge	^2.6.0	Runtime, UI, build, test, or server support library.
Frontend dependency	tailwindcss-animate	^1.0.7	Runtime, UI, build, test, or server support library.
Frontend dependency	vaul	^0.9.9	Runtime, UI, build, test, or server support library.
Frontend dependency	zod	^3.25.76	Runtime, UI, build, test, or server support library.
Frontend dev dependency	@eslint/js	^9.32.0	Runtime, UI, build, test, or server support library.
Frontend dev dependency	@tailwindcss/typography	^0.5.16	Runtime, UI, build, test, or server support library.
Frontend dev dependency	@testing-library/jest-dom	^6.6.0	Runtime, UI, build, test, or server support library.
Frontend dev dependency	@testing-library/react	^16.0.0	Runtime, UI, build, test, or server support library.
Frontend dev dependency	@types/node	^22.16.5	Runtime, UI, build, test, or server support library.
Frontend dev dependency	@types/react	^18.3.23	Runtime, UI, build, test, or server support library.
Frontend dev dependency	@types/react-dom	^18.3.7	Runtime, UI, build, test, or server support library.
Frontend dev dependency	@vitejs/plugin-react-swc	^3.11.0	Runtime, UI, build, test, or server support library.
Frontend dev dependency	autoprefixer	^10.4.21	Runtime, UI, build, test, or server support library.
Frontend dev dependency	eslint	^9.32.0	Runtime, UI, build, test, or server support library.
Frontend dev dependency	eslint-plugin-react-hooks	^5.2.0	Runtime, UI, build, test, or server support library.
Frontend dev dependency	eslint-plugin-react-refresh	^0.4.20	Runtime, UI, build, test, or server support library.
Frontend dev dependency	globals	^15.15.0	Runtime, UI, build, test, or server support library.
Frontend dev dependency	jsdom	^20.0.3	Runtime, UI, build, test, or server support library.
Frontend dev dependency	postcss	^8.5.6	Runtime, UI, build, test, or server support library.
Frontend dev dependency	tailwindcss	^3.4.17	Utility-first styling system.
Frontend dev dependency	typescript	^5.8.3	Static type checking.
Frontend dev dependency	typescript-eslint	^8.38.0	Runtime, UI, build, test, or server support library.
Frontend dev dependency	vite	^5.4.19	Fast local dev server and Rollup production build.
Frontend dev dependency	vitest	^3.2.4	Unit test runner.
Functions dependency	@google/generative-ai	^0.24.1	Server SDK for Gemini PDF extraction.
Functions dependency	firebase-admin	^12.0.0	Admin SDK for privileged Cloud Function database access.
Functions dependency	firebase-functions	^5.0.0	Firebase serverless function framework.
Functions dependency	nodemailer	^8.0.1	SMTP email delivery for contact form notifications.
Functions dev dependency	@types/node	^20.0.0	Runtime, UI, build, test, or server support library.
Functions dev dependency	@types/nodemailer	^7.0.11	Runtime, UI, build, test, or server support library.
Functions dev dependency	typescript	^5.3.3	Static type checking.
Legacy source reference	@supabase/supabase-js	not in current package.json	Generated Supabase client exists in src/integrations/supabase but active runtime uses Firebase.

Additional runtime/deployment tools include Firebase CLI, Firebase Emulator Suite, Hostinger static hosting/CDN, Google Cloud Secret Manager for function secrets, and optional FTP/File Manager deployment for the static dist output.

8. All External APIs & Services Used

Service	Purpose	Calls / Methods	Data Sent / Received	Auth / Limits / Cost
Google Gemini API	Primary AI generation and JSON analysis.	POST generativelanguage.googleapis.com/v1beta/openai/chat/completions; @google/generative-ai generateContent.	Prompts, resume/profile/JD/project/PDF data; receives JSON text, usage metadata, extracted text.	Server-only GEMINI_API_KEY. Estimated cost constants: USD 0.075/M input tokens and USD 0.30/M output tokens.
Firebase Auth	User identity and sessions.	createUserWithEmailAndPassword, signInWithEmailAndPassword, signInWithPopup, sendEmailVerification, sendPasswordResetEmail, onAuthStateChanged, reload, signOut.	Email/password, display name, OAuth provider state; receives Firebase User and token state.	Client Firebase config plus Firebase-managed tokens. Authorized domains required.
Firestore	Database.	Client getDoc/getDocs/query; Admin SDK collection/doc/batch/runTransaction.	Plan documents, analysis results, usage logs, contact forms.	Client constrained by firestore.rules; Admin SDK bypasses rules inside Functions.

Firebase Cloud Functions	Serverless backend.	httpsCallable from frontend; onCall and Auth onCreate on backend.	Structured JSON request/response; request.auth included by SDK.	Firebase callable auth context. Function timeouts 60-120s.
Hostinger SMTP	Contact form email.	nodemailer smtp.hostinger.com:465 sendMail.	Contact name/email/subject/message to hello@aivilabs.com.	SMTP_PASSWORD server env. If absent, message still saves.
Google Fonts	Typography assets.	CSS @import from fonts.googleapis.com.	Browser fetches font CSS/assets.	Public unauthenticated asset requests.
Google Analytics gtag	Optional analytics events.	window.gtag('event', ...).	Event names and small metadata.	Only called if gtag exists on window.
Supabase	Legacy migration/client artifacts.	Generated client and Deno Edge Functions exist; active code primarily uses Firebase.	Legacy schema includes profiles/resumes/analysis/user_plans.	Not active in current runtime; verify before deleting.
Lovable AI Gateway	Legacy AI proxy.	POST ai.gateway.lovable.dev/v1/chat/completions.	Legacy prompts and PDF payloads.	LOVABLE_API_KEY in legacy functions.
Razorpay / Stripe	Planned billing integration.	No live code path, SDK, order creation, or webhook handler currently present.	Not applicable today.	Future secrets and webhook signature verification required.

9. Internal API Reference

Current internal APIs are Firebase callable functions, not REST routes. The browser uses Firebase Functions SDK, which sends authenticated POST requests to deployed callable endpoints. Conventional /api/* routes are not defined by this app; static hosting catch-all rewrites can return the SPA shell for such paths.

Method	Path / Callable	Description	Auth Required	Request Body	Response Schema	Rate Limit
Auth trigger	onUserSignupComplete	Seeds profiles, user_plans, and user_roles for new Firebase Auth users.	Firebase system	Firebase User	No direct response	N/A
POST	applyCoupon	Redeems FIRST1000 Starter trial.	Yes	{ code }	{ success, message }	1000 global; one trial per account
POST	checkUsage	UI quota preflight.	Yes	{ analysisType }	{ allowed, plan, used, limit, remaining, reason }	None
POST	getUserPlanInfo	Plan and usage fetch.	Yes	{}	PlanInfo fields	None
POST	careerAnalysis	Main AI engine for resume, job_fit, roadmap, interview, promotion, portfolio, resume_rebuild.	Yes	{ analysis_type, resume_text?, job_description?, projects?, current_role?, target_role?, missing_skills? }	{ result, cached, content_hash }	Plan quota; 120s timeout
POST	saveAnalysisResult	Stores result/scores from frontend.	Yes	{ analysisType, result, scores, contentHash? }	{ id }	None
POST	extractPdf	Extracts text from PDF base64 with Gemini multimodal.	Yes	{ pdf_base64 }	{ text }	Plan quota; 60s timeout
POST	adminUpdateUser	Admin-only role/plan/quota update.	Yes, admin	{ targetUserId, role?, planType?, resetQuota? }	{ success }	None
POST	linkedinAnalyze	Analyzes LinkedIn text.	Yes	{ profileText }	{ analysisId, result, cached }	3/min/user
POST	linkedinOptimize	Generates optimized LinkedIn profile copy.	Yes	{ analysisId, targetRole?, industry? }	{ optimizeId, result }	3/min/user
POST	submitContactForm	Stores and emails contact form.	No explicit auth	{ name, email, subject, message }	{ success, message }	Length validation only

10. Database Schema

The live database is Firestore. Existing Supabase migrations define a PostgreSQL schema from an earlier migration path and should be treated as legacy unless reactivated.

Model	Storage	Fields	Constraints / Relationships / Indexes
profiles	Firestore	userId, fullName, email, createdAt, updatedAt	Document ID equals Firebase UID; users own own profile; admins can

			read all.
user_plans	Firestore	userId, planType/plan_type, monthly_analysis_used, monthly_reset_date, lifetime counters, trial, trialExpiresAt, trialUsed, couponCode	Document ID equals UID; client read only; server/admin writes only.
user_roles	Firestore	userId, role	Document ID equals UID; role user/admin controls admin panel.
analysis_results	Firestore	userId, analysisType, result, scores, contentHash, parentAnalysisId, createdAt	Stores all AI outputs and cached results.
ai_usage_logs	Firestore	userId, analysisType, model, tokensUsed, promptTokens, completionTokens, estimatedCost, cached, createdAt	Financial/operational audit log.
system_stats/beta	Firestore singleton	first1000Used, updatedAt	Global FIRST1000 trial redemption counter.
ai_rate_limits	Firestore	count, resetAt	Per-user per-function limiter for LinkedIn functions.
contact_messages	Firestore	name, email, subject, message, status, createdAt	Contact form storage for admin review.
resumes	Firestore	userId, resume fields	Rules exist; active code mainly stores analysis_results.
job_descriptions	Firestore	userId, JD fields	Rules exist for future/direct CRUD patterns.
Supabase public tables	Legacy migrations	profiles, resumes, job_descriptions, analysis_results, ai_usage_logs, user_plans, user_roles	Present in migrations and docs; current runtime has moved to Firebase.

10.1 Firestore Indexes

Collection	Fields	Purpose
analysis_results	userId ASC, analysisType ASC, createdAt DESC	Fast per-user history by module.
analysis_results	analysisType ASC, contentHash ASC, createdAt DESC	Global job-fit dedup lookup by hash.
ai_usage_logs	userId ASC, analysisType ASC, createdAt DESC	Usage analytics and history queries.

10.2 Migration Status

Supabase migrations and functions remain in the repository, but the active runtime imports Firebase clients and Firebase callable functions. Treat Supabase tables/RLS/RPC definitions as migration history and reference material. Before removing them, verify no deployed environment still uses the Supabase project in supabase/config.toml.

11. Authentication & Authorisation

Authentication is implemented with Firebase Auth. The frontend initializes Firebase from VITE_FIREBASE_* variables, sets browserLocalPersistence, and uses onAuthStateChanged to keep user state in AuthContext. The supported sign-in methods in the UI are email/password and Google popup. LinkedIn sign-in is shown as disabled/coming soon.

Concern	Implementation
Sign up	createUserWithEmailAndPassword; optional updateProfile(displayName); sendEmailVerification; backend onUserSignupComplete seeds Firestore records.
Sign in	signInWithEmailAndPassword or signInWithPopup(new GoogleAuthProvider()).
Session persistence	browserLocalPersistence via setPersistence(auth, browserLocalPersistence).
Protected routes	ProtectedRoute checks AuthContext user and redirects unauthenticated visitors to /auth.
Roles	user_roles/{uid}.role stores user/admin. AuthContext reads it and AdminDashboard verifies again through adminUpdateUser.
Token validation	Firebase callable Functions receive request.auth automatically; functions reject unauthenticated calls.
Email verification	Dashboard shows warning and resend/refresh controls; verification is encouraged but not hard-blocking every dashboard route.

Password reset	sendPasswordResetEmail from Auth page.

12. Rate Limiting & Security

Control	Current State
Plan quotas	checkUsageInline runs in Firestore transactions; free lifetime limits, Starter 30/month, Pro/admin effectively unlimited.
Per-user rate limit	checkRateLimit stores ai_rate_limits/{uid_functionName}; currently used by linkedinAnalyze and linkedinOptimize at 3/min/user.
Auth checks	Most callables require request.auth. submitContactForm allows unauthenticated submissions with field length validation.
Owner checks	linkedinOptimize loads parent analysis and rejects if analysisData.userId does not match request.auth.uid.
Firestore rules	Client reads/writes restricted by owner/admin; user_plans and analysis_results writes are server-only.
Input validation	Basic presence checks, analysis type validation, PDF presence, contact field length caps, resume upload type/size checks.
CORS	Firebase callable functions handle SDK CORS; legacy Supabase Edge Functions use Access-Control-Allow-Origin: *.
Security headers	No explicit production header config in repo. Recent live check showed missing anti-clickjacking headers; add X-Frame-Options or CSP frame-ancestors.
Secrets	GEMINI_API_KEY and SMTP_PASSWORD are server-only. service-account.json is tracked and must be removed/rotated.
Payment safety	Not applicable today: Razorpay/Stripe webhooks are not implemented.

13. Background Jobs & Workers

Not applicable: the current project does not define a queue, cron job, scheduled function, durable worker, or background job runner. All server-side work is request-triggered through Firebase callable functions or the Firebase Auth onCreate trigger.

Asynchronous behavior still exists inside request lifecycles: Firestore transactions, Gemini API calls, email sending, and fire-and-forget system event logging. Future docs recommend a scheduled analytics aggregation job for admin dashboards, but it is not implemented in the current codebase.

14. Environment Variables

Variable Name	Description	Example Value	Required
VITE_FIREBASE_API_KEY	Firebase browser API key for client SDK initialization.	AIzaSy...	Yes
VITE_FIREBASE_AUTH_DOMAIN	Firebase Auth domain.	project.firebaseapp.com	Yes
VITE_FIREBASE_PROJECT_ID	Firebase project ID.	aivi-career	Yes
VITE_FIREBASE_STORAGE_BUCKET	Firebase storage bucket identifier.	project.appspot.com	Yes
VITE_FIREBASE_MESSAGING_SENDER_ID	Firebase messaging sender ID.	1234567890	Yes
VITE_FIREBASE_APP_ID	Firebase app ID.	1:123:web:abc	Yes
VITE_USE_FIREBASE_EMULATOR	Connects SDKs to local emulators when true.	true	No
GEMINI_API_KEY	Server-only Gemini API key used by Firebase Functions.	AIza...	Yes for AI/PDF
SMTP_PASSWORD	Server-only password for Hostinger SMTP contact emails.	app-password	No
VITE_SUPABASE_URL	Legacy Supabase client URL still referenced by generated client.	https://project.supabase.co	No for Firebase runtime
VITE_SUPABASE_PUBLISHABLE_KEY	Legacy Supabase anon/publishable key.	eyJ...	No for Firebase runtime
VITE_SUPABASE_PROJECT_ID	Legacy Supabase project reference from migration docs.	project-ref	No

SUPABASE_URL	Legacy Supabase Edge Function server URL.	https://project.supabase.co	Legacy only
SUPABASE_SERVICE_ROLE_KEY	Legacy Supabase server-only key.	eyJ...	Legacy only
LOVABLE_API_KEY	Legacy Lovable AI Gateway key in Supabase Edge Functions.	lvb_...	Legacy only
FIRESTORE_EMULATOR_HOST	Local admin scripts target Firestore emulator.	localhost:8080	Scripts only
FIREBASE_AUTH_EMULATOR_HOST	Local admin scripts target Auth emulator.	localhost:9099	Scripts only

14.1 .env.example

```
# Frontend Firebase config
VITE_FIREBASE_API_KEY=
VITE_FIREBASE_AUTH_DOMAIN=
VITE_FIREBASE_PROJECT_ID=
VITE_FIREBASE_STORAGE_BUCKET=
VITE_FIREBASE_MESSAGING_SENDER_ID=
VITE_FIREBASE_APP_ID=
VITE_USE_FIREBASE_EMULATOR=false

# Legacy Supabase client values
VITE_SUPABASE_URL=
VITE_SUPABASE_PUBLISHABLE_KEY=
VITE_SUPABASE_PROJECT_ID=

# Firebase Functions secrets / environment
GEMINI_API_KEY=
SMTP_PASSWORD=

# Legacy Supabase Edge Function secrets
SUPABASE_URL=
SUPABASE_SERVICE_ROLE_KEY=
LOVABLE_API_KEY=

# Local emulator helpers for admin scripts
FIRESTORE_EMULATOR_HOST=localhost:8080
FIREBASE_AUTH_EMULATOR_HOST=localhost:9099
```

15. Local Setup Guide

23. Install Node.js 20 or newer, npm, Git, and Firebase CLI.
24. Clone the repository and enter the project directory.
25. Install frontend dependencies at the repo root.
26. Install Firebase Functions dependencies in functions/.
27. Create .env.local with the VITE_FIREBASE_* values and optional emulator flag.
28. Set Firebase Functions secrets for AI and SMTP if running/deploying functions.
29. Start emulators for Auth, Firestore, Functions, and Hosting when testing locally.
30. Start the Vite dev server and open the local URL.

```
git clone <repo-url> aivi-career
cd aivi-career
npm install
cd functions
npm install
cd ..
```

```
# Create .env.local at repo root, then add VITE_FIREBASE_* values.
# Optional local emulator mode:
# VITE_USE_FIREBASE_EMULATOR=true

# Configure function secrets for deployed functions:
firebase functions:secrets:set GEMINI_API_KEY
firebase functions:secrets:set SMTP_PASSWORD

# Build functions and run emulators:
cd functions
npm run build
cd ..
firebase emulators:start

# In another terminal:
npm run dev
# Verify frontend: http://localhost:5173
# Emulator UI: http://localhost:4000
```

Database setup for the active Firebase path is driven by Firestore security rules, indexes, the Auth onCreate function, and emulator state. Supabase migrations are not required for the current Firebase runtime unless reactivating legacy Supabase paths.

16. Testing

Command	Purpose
npm run test	Runs Vitest tests matching src/**/*.{test,spec}.{ts,tsx}.
npm run test:watch	Runs Vitest in watch mode.
npm run lint	Runs ESLint across the repo.
npm run build	Builds production frontend bundle.
cd functions && npm run build	Compiles Firebase Functions TypeScript.
npx vitest run --coverage	Coverage check if coverage provider is installed/configured; no dedicated coverage script exists today.

Test File	Coverage
src/test/setup.ts	Vitest/jsdom test setup including testing-library jest-dom matchers.
src/test/example.test.ts	Minimal example test; does not yet cover product flows.

Current automated test coverage is minimal. High-value tests to add include jobFitScoring formula tests, Auth route guard tests, planService callable behavior mocks, ResumeUploadForm file validation, UpgradeModal behavior, and Firebase Function unit/integration tests with emulator-backed Firestore.

17. Deployment Guide

The repository supports Firebase deployment configuration, while existing docs describe Hostinger static hosting for the frontend and Firebase Cloud Functions for backend compute. Confirm the target production host before deployment.

```
# Frontend production build
npm run build
```

```
# Firebase Functions build and deploy
cd functions
npm run build
firebase deploy --only functions
cd ..

# Firestore rules and indexes
firebase deploy --only firestore:rules
firebase deploy --only firestore:indexes

# If using Firebase Hosting instead of Hostinger static upload
firebase deploy --only hosting
```

Deployment Area	Steps / Requirements
Frontend static hosting	Build dist/ with npm run build. Upload dist contents to Hostinger public_html or deploy Firebase Hosting.
SPA rewrites	Hostinger requires .htaccess rewrite to index.html. Firebase hosting currently rewrites source ** to /index.html.
Backend	Deploy functions from functions/ after npm run build; Node 20 runtime declared.
Secrets	Set GEMINI_API_KEY and SMTP_PASSWORD using Firebase Secret Manager or environment configuration before deploy.
Database	Deploy firestore.rules and firestore.indexes.json. No active SQL migration is required for Firebase runtime.
Reverse proxy	Not applicable for static Hostinger/Firebase Hosting deployment. If self-hosting, use nginx/caddy only as a static file server with SPA fallback.
SSL	Use Hostinger/Firebase managed SSL. Verify HTTPS redirects and add production security headers.
Docker	Not applicable: no Dockerfile exists in the repository.
Post-deploy verification	Open /, /auth, /dashboard, one public SEO route, and one callable module with a test account. Verify Firestore writes, Gemini response, and contact form save/email behavior.

18. Known Issues & TODOs

Issue	Evidence	Impact	Recommended Action
Tracked service account file	service-account.json is present and tracked by git.	Potential credential exposure if pushed or shared.	Remove from Git history if exposed, add service-account*.json to .gitignore, rotate key, use secrets/local-only credentials.
Payment integration pending	UpgradeModal links to /#pricing; docs state Razorpay/Stripe are stubbed/not live; no SDK/order/webhook code found.	Users cannot self-serve paid upgrades; admin/manual plan changes only.	Implement create order, checkout, webhook signature verification, subscription state sync, and tests.
Security headers missing from repo config	No header config in firebase.json; live check previously showed missing X-Frame-Options/frame-ancestors.	Clickjacking hardening gap and scanner finding.	Add CSP frame-ancestors or X-Frame-Options plus HSTS, referrer, content-type options at host/CDN.
/api/* SPA fallback noise	firebase.json rewrites ** to /index.html; no REST API routes exist.	Scanners may misclassify /api/auth/login or /api/payments as live APIs returning 200.	Add explicit 404/deny handling for /api/** or only rewrite non-API paths.
Supabase legacy residue	supabase client, migrations, docs, and Edge Functions remain but active app uses Firebase.	Confusing onboarding and documentation drift.	Decide whether to archive/remove legacy path or mark it clearly as deprecated.
Docs drift on model names and schema	Some docs mention gemini-1.5-flash and betaExpiry/count; code uses gemini-2.5-flash, trialExpiresAt, first1000Used.	New developers may implement against stale fields.	Treat this document and functions/src/index.ts as current source of truth; update old docs.
PDF extraction consumes resume quota	extractPdf calls checkUsageInline(userId, 'resume') before analysis.	A PDF-based resume analysis can consume quota for extraction and again for analysis.	Separate pdf_extraction from resume analysis quota or make extraction free with separate abuse limit.
Limited automated tests	Only src/test/example.test.ts exists.	Regressions in auth, quotas, prompts, and scoring may go undetected.	Add Vitest coverage and Firebase Functions emulator tests.
Admin analytics scalability	Docs warn raw reads over ai_usage_logs could become expensive at scale.	Large logs can create Firestore cost and browser memory pressure.	Add scheduled aggregation to analytics/daily and paginate details.
Contact form lacks explicit rate limit	submitContactForm has length validation but no user/IP throttle.	Public spam risk.	Add App Check, CAPTCHA, or Firestore/IP based throttling.

19. Glossary

Term	Definition
AIVI	Artificial Intelligence Visibility Insights; the product/brand for career intelligence.
Career Intelligence OS	One workspace for resume, job fit, LinkedIn, skills, interview, promotion, and portfolio intelligence.
ATS	Applicant Tracking System; software used to parse and rank resumes.
Job Fit	Comparison between a user's resume and a target job description.
Resume Intelligence	AIVI module for resume scoring and improvement guidance.
Resume Studio	AIVI module for rebuilding or tailoring resumes.
FIRST1000	Beta coupon granting 30 days of Starter access, capped at 1000 redemptions.
Free plan	Default plan with small lifetime analysis limits and no premium generation features.
Starter plan	Plan with 30 analyses per month; currently activated through coupon/manual update, not live payment.
Pro plan	Unlimited analysis tier in gating logic.
Callable Function	Firebase HTTPS function invoked via Firebase Functions SDK with request.auth context.
Firestore Admin SDK	Privileged server SDK used in Cloud Functions; bypasses client security rules.
Content hash	SHA-256 hash used to identify duplicate job-fit or LinkedIn inputs for cache reuse.
LLM	Large Language Model; Gemini is the LLM provider used by the active backend.
RLS	Row-Level Security; Supabase/PostgreSQL access control present in legacy migrations.
SPA	Single Page Application; React handles routes client-side.
shadcn/ui	Reusable component system built on Radix UI primitives and Tailwind.
Gemini	Google AI model family used for analysis, generation, and PDF extraction.
Hostinger	Static hosting/CDN and SMTP provider referenced by deployment and contact email paths.